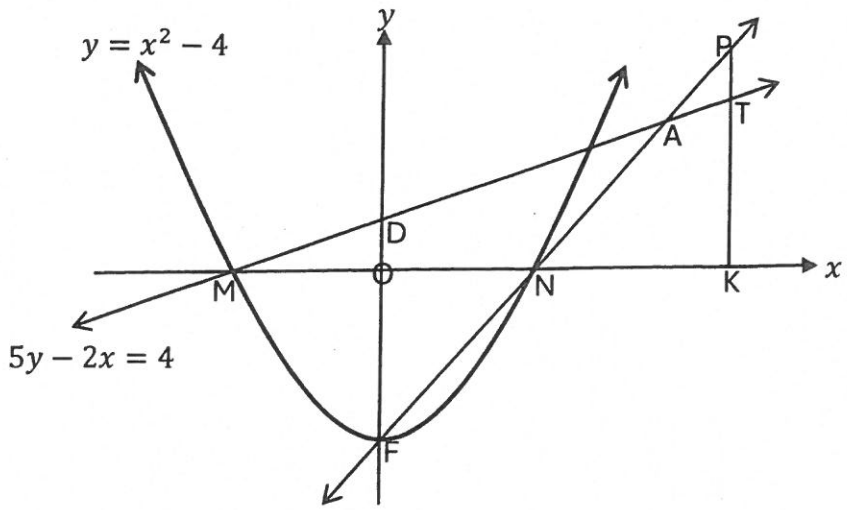
QUESTION 1

1.1 On the same set of axes, draw the graphs of:

$$f(x) = 2^x \; ; g(x) = 2^{-x} \text{ and } h(x) = -2^x \qquad (6)$$

1.2 Determine the equation of the straight line MN if MN $\perp$ CD and with equation $y = -3x - 5$ and it passes through the point N(-6 ; 4). (4)

1.3



Determine:

1.3.1 The equation of line FNP. (5)

1.3.2 The coordinates of A. (4)

1.3.3 The length of DF. (2)

1.3.4 OK if PT = $3\frac{1}{5}$. (4)

1.3.5 x if $x - 2 \geq \frac{x+2}{5}$. (2)

1.3.6 x if $(x^2 - 4)(2x - 4) \leq 0$. (3)

1.3.7 For which values of x is the gradient of the parabola positive? (1)

QUESTION 2

2.1 The following functions are given:

$$f(x) = 2x^2 + 3x - 10 \text{ and } h(x) = x^2 - 3x.$$

Determine:

2.1.1 $f(-2)$ (2)

2.1.2 $h(2)$ (2)

2.2 Sketch the graphs of the following two functions on the set of axes on the last page:

$$p(x) = x + 2 \text{ and } f(x) = 2x^2 - 8$$

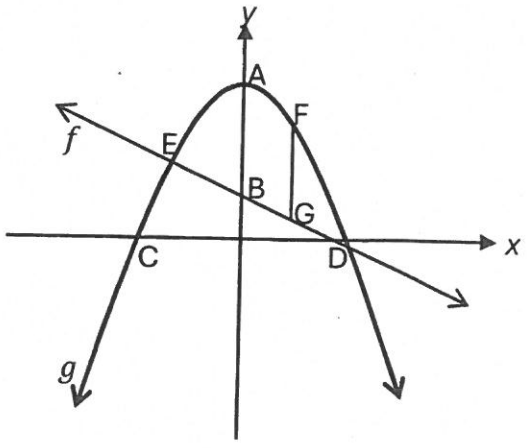
(5)

2.3 Sketch the graphs of the following two functions on the set of axes on the last page:

$$f(x) = \frac{9}{x} + 4 \text{ and } h(x) = -2(2)^x + 2$$

(6)

2.4 The graphs of $f(x) = -2x + 4$ and $g(x) = -2x^2 + 8$ are drawn below:



2.4.1 Determine the coordinates of A, B, C and D. (5)

2.4.2 Determine the coordinates of E, a point where the two graphs intersect. (4)

2.4.3 Determine the distance between C and D. (1)

QUESTION 3

Draw sketch graphs of $p(x) = \sin x$ and $h(x) = \cos x$ on the set of axes on the last page.

(6)

3.1 What is the amplitude of p? (1)

3.2 What is the period of h? (1)

3.3 What is the range of p? (2)

3.4 Give the range of h. (2)

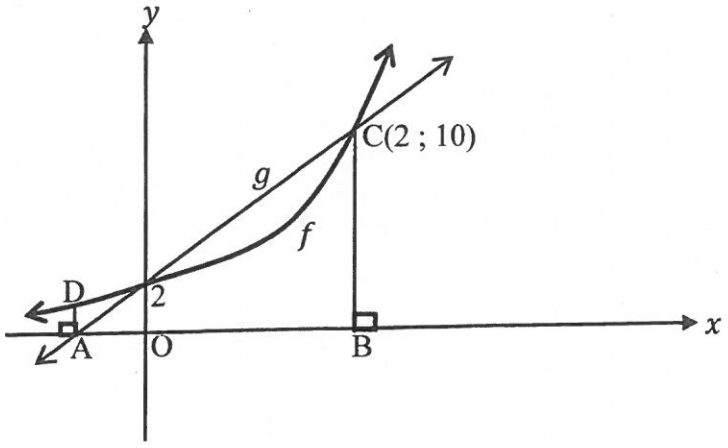
[12]

QUESTION 4

4.1 Draw sketch graphs of $f(x) = -x^2 + 4$ and $g(x) = \frac{6}{x} - 3$ on the same set of axes. Indicate asymptotes and intercepts clearly.

(8)

4.2 The graphs of $f(x) = a^x + q$ and $g(x) = mx + c$ are given.



4.2.1 Determine the equations of f and g.

(7)

4.2.2 Write down the equation of the asymptote of f.

(1)

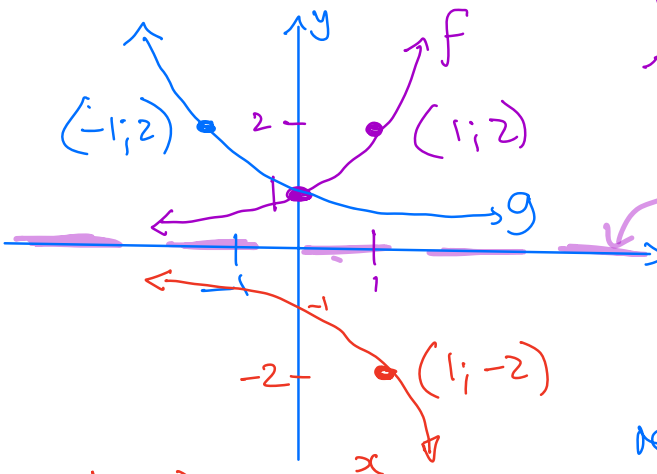
4.2.3 Determine the lengths of OA, AB and BC.

(5)

4.2.4 Hence, determine the lengths of AC and AD.

(4)

Solutions



$f(x) = 2^x$

Asympt $y=0$

Make sure we see it

0	1
$2^0=1$	2

$h(x) = -2^x$

reflect over x axis

$y \to -y$

$g(x) = 2^{-x}$

reflect over y axis

or $\left(\frac{1}{2}\right)^x$ $x \to -x$

1.2 $y = -3x - 5$

$m = -3$

$MN \perp CD$ N $(-6;4)$

$\therefore m = \frac{1}{3}$

$y - 4 = \frac{1}{3}(x+6)$

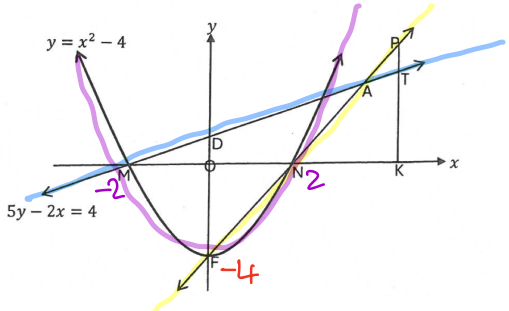
$y = \frac{1}{3}x + 2 + 4$

$= \frac{1}{3}x + 6$

Or $3y - 12 = x + 6$

$3y = x + 18$

1.3



Fill in as you go like geometry

from $y = x^2 - 4$

$0 = (x-2)(x+2)$

$M(-2;0)$ $N(2;0)$

$F(0;-4)$

1.3.1 FNP $y = mx - 4$

$0 = m(2) - 4$

$4 = 2m$

$2 = m$

$\boxed{y = 2x - 4}$

OR

$m = \frac{0 - -4}{2 - 0}$

$m = \frac{4}{2} = 2$

1.3.2. $y = 2x - 4$ $\quad 5y - 2x = 4$

$5y = 2x + 4$

Solve simult

$5(2x - 4) = 2x + 4$

$10x - 20 = 2x + 4$

$8x = 24$

$x = 3$

$A(3; 2)$

OR eliminate

1.3.3 DF

$F(0; -4)$ D is y-int of line

$5y - 2(0) = 4$

$5y = 4$

$y = \frac{4}{5}$

$DF = 4 + \frac{4}{5}$

$= 4\frac{4}{5} = \frac{24}{5}$ u.

1.3.4 $PT = top - bottom$

$3\frac{1}{5} = 2x - 4 - \left(\frac{2x+4}{5}\right)$

$\frac{16}{5} = 2x - 4 - \left(\frac{2x+4}{5}\right)$

$16 = 10x - 20 - 2x - 4$

$80 = 8x$

$5 = x$

OK = 5 units

You can also be told OK = 5 find PT

1.35. $x - 2 \geq \frac{x+2}{5}$

$5x - 10 \geq x + 2$

$4x \geq 12$

$x \geq 3$

1.3.6. $(x^2 - 4)(2x - 4) \leq 0$

Will be told to use your graphs to find x

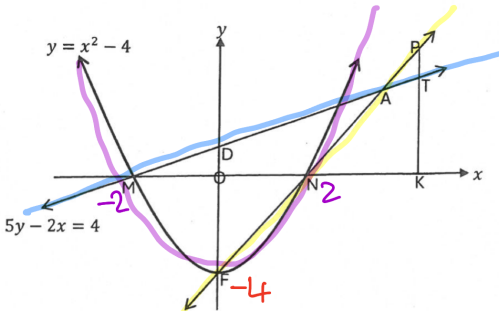
(parab)(2x-4) ≤ 0

one pos one neg

or

either/both 0

$x \leq -2$ ✓ and $x = 2$ ✓



1.3.7 $x \geqslant 0$ ✓ pos m

2.1 $f(x) = 2x^2 + 3x - 10$ $h(x) = x^2 - 3$

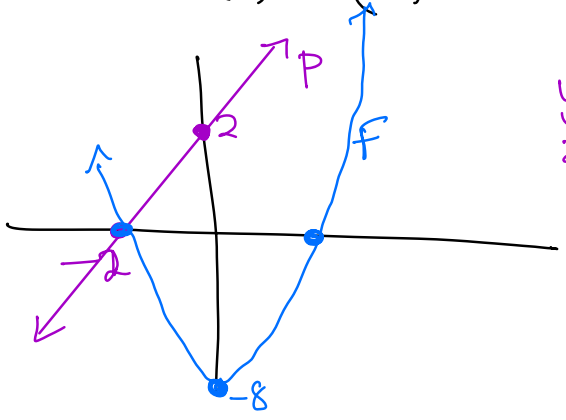
2.1.1 $f(-2) = 2(-2)^2 + 3(-2) - 10$

$= 8 - 6 - 10$

$= -8$

2.1.2 $h(2) = (2)^2 - 3 = 1$

2.2.



$y = x + 2$

$x = 0 \quad y = 2$

$y = 0 \quad x = -2$

$f(x) = 2x^2 - 8$

y int $(0; -8)$

$0 = 2x^2 - 8$

$0 = x^2 - 4$

$0 = (x-2)(x+2)$

$(-2; 0)$

$(2; 0)$

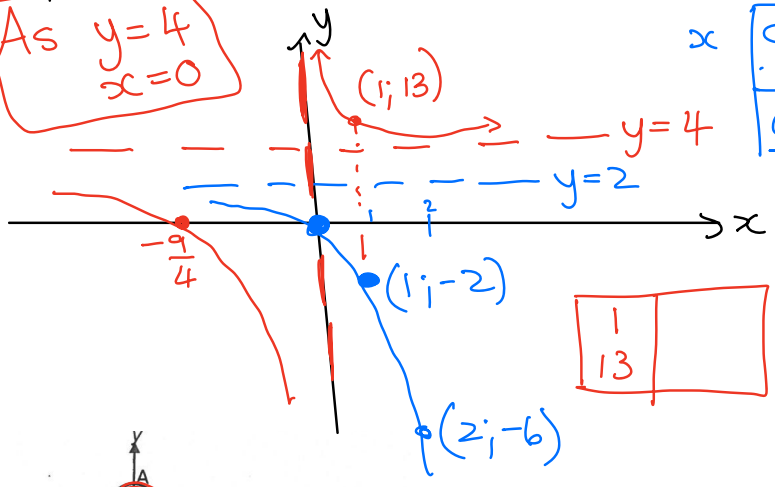
2.3 $f(x) = \frac{9}{x} + 4$ $h(x) = -2(2^x) + 2$.

$\frac{9}{x} = -4$

$\frac{9}{-4} = x$

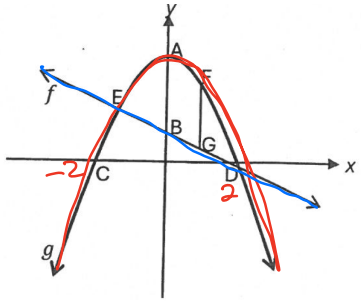
As $y = 4$, $x = 0$

x	0	1	2
	0	-2	-6



1	
13	

2.4



$g(x) = -2x^2 + 8$

A (0; 8)

C, D $-2x^2 + 8 = 0$

$-2x^2 = -8$

$x^2 = 4$

$x = \pm 2$

$f(x) = -2x + 4$

B (0; 4).

C (-2; 0)

D (2; 0)

2.4.2 $-2x + 4 = -2x^2 + 8$

$2x^2 - 2x - 4 = 0$

$x^2 - x - 2 = 0$

D $(x - 2)(x + 1) = 0$

$x = 2$ or $x = -1$

Look at picture to check factors.

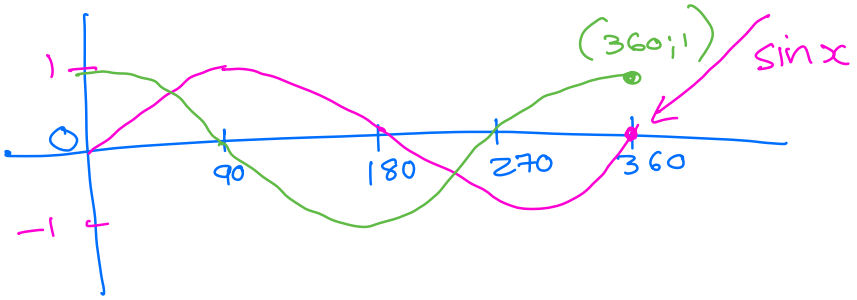
E (-1; 6) we already know D

2.4.3 C (-2; 0) D (2; 0)

CD = 4 units

3

Paper 2



3.1 amp = 1 3.2 period = 360°

3.3 range $y \in [-1;1]$ or $-1 \leq y \leq 1$

3.4 range $y \in [-1;1]$

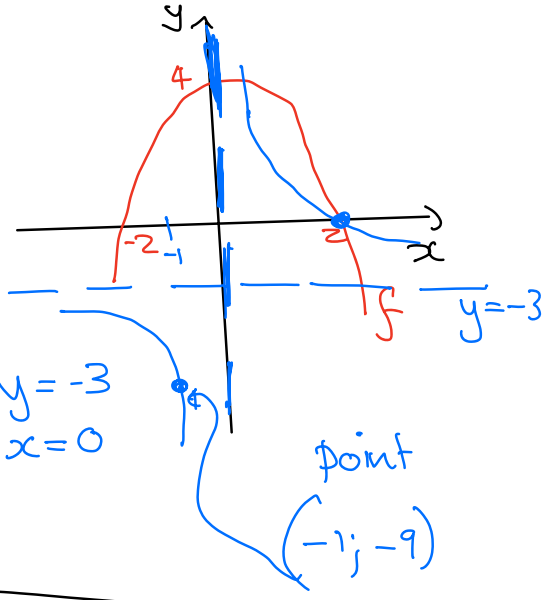
4.1 $f(x) = -x^2 + 4$

y int $(0;4)$

x int $x^2 - 4 = 0$

$(x+2)(x-2) = 0$

$(-2;0)$ $(2;0)$



$g(x) = \frac{6}{x} - 3$

As $y = -3$ $x = 0$

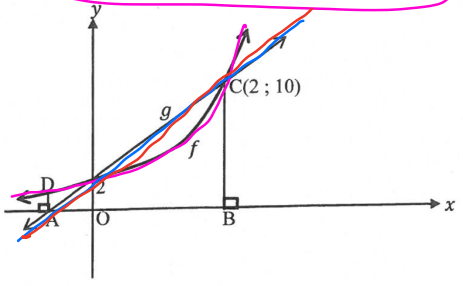
point $(-1;-9)$

x int

$\frac{6}{3} - x = 0$

$2 = x$

4.2 $f(x) = a^x + q$ $g(x) = mx + c$



line $c = 2$

$y = mx + 2$

$10 = m(2) + 2$

$8 = 2m$

$4 = m$

$g(x) = 4x + 2$

$f(x) = a^x + q$

$2 = a^0 + q$

$1 = q$

$10 = a^2 + q$

$10 = a^2 + 1$

$9 = a^2$

$\pm 3 = a$ But incr graph

$\therefore a = 3$

$f(x) = 3^x + 1$

As of f $y = 1$

$OA = \frac{1}{2}$ u

$4x + 2 = 0$

$4x = -2$

$x = -\frac{1}{2}$

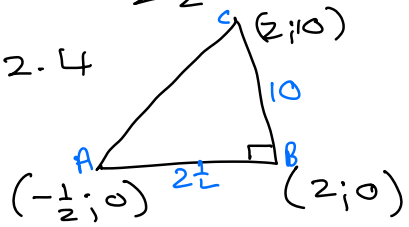
4.2.3 $AB = AO + OB$

$= \frac{1}{2} + 2$

$= 2\frac{1}{2}$ units

$BC = 10$ units

4.2.4



$(2\frac{1}{2})^2 + (10)^2 = AC^2$ pythag

$106,25 = AC^2$

$10,31 = AC$

2dp.

AD D$(-\frac{1}{2}; \quad)$

$y = 3^{-\frac{1}{2}} + 1$

$= \sqrt{\frac{1}{3}} + 1$

$= 1,577$ units